One reversible measure, but a non-reversible stationary measure: exclusion processes

counterexamples-ips, entry 0001

Status: *plausible* (not yet checked by a human)

Idea and write-up: Claude (Opus 5.5), 2 November 2026

Question. Let an interacting particle system on $\{0,1\}^{\mathbb{Z}^d}$ have translation-invariant, finite-range rates. If it has at least one reversible measure, is every stationary measure reversible?

For stochastic Ising models this is the Holley–Stroock question: true for translation-invariant stationary measures in all d (Holley 1971), true for all stationary measures when $d \le 2$ (Holley–Stroock 1977), open for $d \ge 3$. In the generality above, the answer is *no*.

Setting. Nearest-neighbour asymmetric simple exclusion (ASEP) on $\mathbb{Z}$: particles jump right at rate p and left at rate q, $p > q > 0$, jumps to occupied sites being suppressed:

$$\Omega f(\eta) = \sum_{x\in\mathbb{Z}} \Big(p\,\eta(x)(1-\eta(x+1)) + q\,\eta(x+1)(1-\eta(x)) \Big) \big(f(\eta^{x,x+1}) - f(\eta) \big).$$

Let $\lambda = p/q > 1$.

Proposition 1. *For ASEP on $\mathbb{Z}$ with $p > q$:*

1. *for every $c > 0$, the product measure μ_c with $\mu_c(\eta(x) = 1) = \rho_x := c\lambda^x/(1+c\lambda^x)$ (a* blocking measure*) is reversible;*
2. *for every $\rho \in (0,1)$, the Bernoulli product measure ν_ρ is stationary but not reversible.*

Proof. (1) For product measures and exclusion dynamics it suffices to check detailed balance on each bond (Liggett 1985, Ch. VIII): $p\,\rho_x(1-\rho_{x+1}) = q\,\rho_{x+1}(1-\rho_x)$. Since $\rho_y/(1-\rho_y) = c\lambda^y$, the ratio $\rho_{x+1}(1-\rho_x)/(\rho_x(1-\rho_{x+1}))$ equals $\lambda = p/q$.

(2) Stationarity of ν_ρ holds for every translation-invariant exclusion process (Liggett 1985, Ch. VIII). For non-reversibility, let $f(\eta) = \eta(0)(1-\eta(1))$ and $g(\eta) = (1-\eta(0))\eta(1)$. On $\{f = 1\}$ we have $g = 0$, and the only transition producing $g = 1$ is the jump $0 \to 1$, at rate p; all other transitions touching 0 or 1 are blocked or lead to $(\eta(0),\eta(1)) \in \{(0,0),(1,1)\}$. Hence $f\,\Omega g = pf$, and likewise $g\,\Omega f = qg$, so

$$\int f\,\Omega g\,d\nu_\rho = p\rho(1-\rho) \neq q\rho(1-\rho) = \int g\,\Omega f\,d\nu_\rho. \qquad \square$$

Mechanism. Reversibility means zero net current across every bond. Under ν_ρ the current is $(p-q)\rho(1-\rho) > 0$: a translation-invariant state with drift transports particles. The blocking measures pile the particles up ($\rho_x \to 0$ as $x \to -\infty$, $\rho_x \to 1$ as $x \to +\infty$), so that exclusion exactly balances the drift on every bond. Conservation of particles lets both kinds of stationary state coexist in sectors of the state space that the dynamics never connects.

Remark (Degenerate version). For any exclusion process $\delta_{\mathbf{0}}$ and $\delta_{\mathbf{1}}$ are reversible, so the claim fails for asymmetric exclusion in every dimension. The blocking measures show the failure is not merely due to absorbing configurations.

Remark (Full support, $d \geq 2$; sketch). On $\mathbb{Z}^2$ let particles jump to $x \pm e_1$ at rates p, q ($p > q$) and to $x \pm e_2$ at rate r each. The product measure with $\rho_x/(1-\rho_x) = \lambda^{x_1}$ satisfies detailed balance on every bond, so it is reversible with full support, while ν_ρ is stationary and not reversible.

Which hypothesis breaks. The claim must exclude conservative dynamics. In the Holley–Stroock setting the rates are positive spin-flip rates, so there is no conservation law and no decomposition into sectors.

Follow-up questions. See `entries/0001-asep-blocking-vs-product-measures/followups.md`.

References

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